

PRST Class (08/10/2020)

Weak law of large number (WLLN)

If $\{X_n\}$ is a sequence of independent and identically distributed rvs, with $E(X_i) = \mu$

$$S_n = X_1 + X_2 + \dots + X_n$$

$$\frac{S_n}{n} \xrightarrow{P} \mu$$

$$\frac{S_n}{n} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\begin{aligned} \mu &= E(X_1) \\ &= E(X_2) \\ &\vdots \\ &= E(X_n) \end{aligned}$$

Eg

X_n : denotes the number coming from the n th throw of a die

$\{X_n\}$ are iid

$$\frac{1}{n} \sum_{i=1}^n X_i = \frac{X_1 + X_2 + \dots + X_n}{n}$$

$$\xrightarrow{P} 3.5$$

X_n	$P(X_n = x)$
1	$\frac{1}{6}$
2	$\frac{1}{6}$
3	$\vdots$
4	$\vdots$
5	$\vdots$
6	$\frac{1}{6}$

$$\begin{aligned} E(X_n) &= \sum x_i \frac{1}{6} \\ &= \frac{1}{6} (1+2+3+4+5+6) \\ &= 3.5 \end{aligned}$$

$$\begin{array}{c} \text{1st throw} \quad \text{2nd throw} \quad \text{3rd throw} \quad \vdots \\ \uparrow \quad \uparrow \quad \uparrow \\ 1 + 2 + 4 + 4 + 5 + \dots + 6 \\ \hline 100 \end{array} \rightarrow 3.5$$

law of large number

Strong law

If $\{X_n\}$ is sequence of rvs, independent and identically distributed, with $E(X_n) = \mu$

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} \mu$$

$$\lim_{n \rightarrow \infty} \frac{S_n}{n} = \mu$$

Person 1 — (n) times throwing $\frac{S_n}{n}$
 Person 2
 ...
 Person 1000

→ 3.5

WLLN
SLLN

$$\frac{S_n}{n} \rightarrow 3.5$$

SLLN

Central limit theorem

Let $\{X_n\}$ be a seq of rvs independent and identically distributed
 $E(X_n) = \mu$, $\text{Var}(X_n) = \sigma^2 > 0$

$$\lim_{n \rightarrow \infty} P\left(\frac{\frac{S_n}{n} - \mu}{\frac{\sigma}{\sqrt{n}}} \leq z\right) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du$$

$$X: f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, -\infty < x < \infty$$

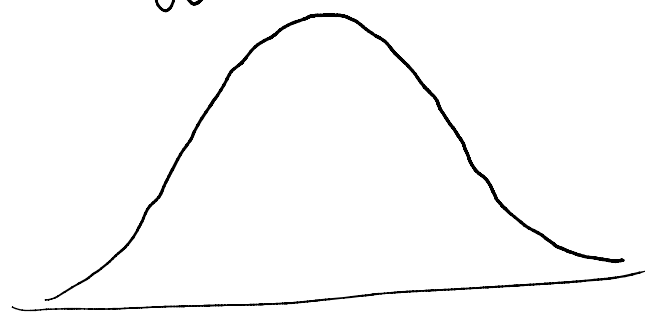
$$X \sim N(0, 1)$$

$$E(X) = 0$$

$$V(X) = 1$$

$$F(x) = \int_{-\infty}^x f(u) du$$

$$= \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du$$



$$Z = \frac{\frac{S_n}{n} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$-\infty < Z < \infty$$

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$

$$P(Z \leq x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} du$$

Ex 1. A computer has a random number generator that generates number between 0 and 1.

Suppose we sum it 100 times and S_{100} be the sum of 100 numbers.

$$P(S_{100} \geq 52) = ? \quad \left| \begin{array}{l} E(X_n) = \frac{1}{2} \\ \text{Var}(X_n) = \frac{1}{12} \end{array} \right.$$

$$\frac{\frac{S_{100}}{100} - E(X_n)}{\frac{\sqrt{\text{Var}(X_n)}}{10}}$$